ModelIt!

Teacher's Guide

When the Earth Cracks Open

The Powerful Forces That Shape Our Planet's Surface

Quick Reference

Grade Level	Grade 6
NGSS Standard	MS-ESS2-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-ESS2-2

Construct an explanation based on evidence for how geoscience processes have changed Earth's surface at varying time and spatial scales.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how mantle heat, plate speed, rock density, and friction interact to produce geological events.
Disciplinary Core Idea	**ESS2.B Plate Tectonics and Large-Scale System Interactions** — The movement of tectonic plates is driven by convection in the mantle and results in continental drift, earthquakes, volcanoes, and mountain building.
Crosscutting Concept	**Stability and Change** — Students explore how Earth's surface appears stable but is constantly changing due to slow, powerful tectonic forces — and how small changes build up to dramatic events.

CAST Testing Alignment

MULTIPLE CHOICE:

Scientists observe that earthquakes along the San Andreas Fault occur in clusters every 100-150 years. Which best explains this pattern using plate tectonic theory?

CONSTRUCTED RESPONSE:

Using your model of plate tectonics, explain why the west coast of South America has many volcanoes and earthquakes while the east coast does not.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Plate tectonics is the grand unifying theory of geology. It explains earthquakes, volcanoes, mountain building, ocean floor spreading, and even the distribution of fossils across continents.

Earth's Layers

Earth consists of the crust (thin outer layer), mantle (thick semi-solid layer), outer core (liquid iron), and inner core (solid iron). The crust and uppermost mantle form the lithosphere, which is broken into tectonic plates that float on the more fluid asthenosphere below.

Convection Drives Plates

Heat from Earth's core and radioactive decay in the mantle creates convection currents — huge circular flows of rock that rise when hot and sink when cool. These currents drag tectonic plates along, causing them to move 2-15 cm per year. This slow movement is responsible for all major geological features.

Types of Plate Boundaries

Divergent boundaries: plates move apart, creating new crust (mid-ocean ridges). Convergent boundaries: plates collide, creating mountains or subduction zones. Transform boundaries: plates slide past each other, creating earthquake faults like the San Andreas. Each type produces different geological hazards.

The Ring of Fire

About 75% of the world's volcanoes and 90% of earthquakes occur along the Ring of Fire — the boundaries of the Pacific Plate. This zone stretches from New Zealand through Japan, across Alaska, and down the west coast of the Americas. It's the most geologically active zone on Earth.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify plate movement speed, mantle heat, rock density, and boundary friction as the key components of the tectonic system.

E **Establish Relationships**
Students determine that mantle heat positively drives plate movement, while boundary friction negatively affects smooth plate motion but positively affects earthquake intensity when released.

V **Visualize & Model**
Students build a model showing how heat-driven plate movement interacts with friction to produce patterns of geological events.

E **Evaluate Outcomes**
Students run scenarios comparing low friction (frequent small quakes) vs. high friction (rare but massive quakes) to understand earthquake patterns.

R **Revise & Extend**
Students add magma pressure or water content to model volcanic activity alongside earthquakes.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"The ground beneath your feet is moving RIGHT NOW. You just can't feel it."*

Watch For

Show a short video of earthquake footage to create urgency and wonder.

Slide 2: Learning Objectives

3 min

Say: *"Today we're going to model the forces that crack open our planet."*

Watch For

Have students read objectives. Pre-teach 'tectonic plate' and 'convection current.'

Slide 3: Big Question

3 min

Say: *"Look at this earthquake map. Do you see a pattern? It's NOT random."*

Watch For

Show global earthquake/volcano map. Have students describe patterns they see.

Slide 4: LEVER Framework

2 min

Say: *"We'll build a model to understand why California gets earthquakes but Florida doesn't."*

Watch For

Preview LEVER steps. Connect to students' earthquake experiences.

Slide 5: Activity 1: Components

8 min

Say: *"What forces are at work beneath our feet? Which ones can we observe?"*

Watch For

Guide sorting. Explain why mantle heat is external (comes from Earth's core).

Slide 6: Activity 2: Connections

8 min

Say: *"Here's the tricky part: MORE friction doesn't mean fewer earthquakes — it means BIGGER ones."*

Watch For

Demonstrate with two erasers sliding past each other. More friction = more buildup = bigger snap.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's crank up the heat and see what happens to our plates!"*

Watch For

Students predict before running. Compare low-friction vs. high-friction boundary scenarios.

Slide 8: Discoveries

5 min

Say: *"Now you know why earthquakes and volcanoes aren't random — they follow the plates."*

Watch For

Connect model results to real Ring of Fire data. Show plate boundary map.

Slide 9: STEM Challenge

12 min

Say: *"You're now engineers. Design a city that can survive what our model predicts."*

Watch For

Distribute materials. Groups design earthquake-resistant city with evidence from model.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Mantle Heat
- **INTERNAL:** Plate Movement Speed, Rock Density, Boundary Friction

Relationships

- **Mantle Heat to Plate Movement Speed:** POSITIVE (+) — Higher mantle heat creates stronger convection currents, which push plates faster. The hotter the mantle, the more energy drives plate motion.
- **Plate Movement Speed to Boundary Friction:** POSITIVE (+) — Faster-moving plates create more friction at boundaries where they meet. More speed means more grinding force.
- **Boundary Friction to Earthquake Intensity:** POSITIVE (+) — More friction at a boundary means more energy builds up before release. When locked plates finally slip, the accumulated energy produces larger, more destructive earthquakes.

Key Discoveries

- Earthquakes and volcanoes cluster along plate boundaries — they're not random at all
- Higher mantle heat drives faster convection currents, pushing plates faster
- More boundary friction means energy builds up longer — resulting in BIGGER earthquakes when it finally releases
- The densest plates get pushed under lighter plates (subduction), creating deep ocean trenches and volcanoes

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The ground is solid and doesn't move"	Earth's surface is made of plates that are constantly moving, just very slowly (a few cm/year). Over millions of years, this movement reshapes continents, builds mountains, and opens oceans. We don't feel it because it's too slow.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Earthquakes happen randomly"	Earthquakes follow tectonic plate boundaries with remarkable precision. 90% of earthquakes occur along the Ring of Fire. If you plot earthquake locations on a map, they trace plate boundaries perfectly.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Volcanoes and earthquakes aren't related"	Both are caused by tectonic plate interactions. At convergent boundaries, subduction creates both deep earthquakes and volcanic chains. The same forces that cause one cause the other.